SAMPLE ACTUARIAL ANALYTICS DIAGNOSTIC

P&C Pricing Diagnostic

A compact review that helps insurance teams identify segments with elevated claim frequency, loss ratio pressure, calibration issues, and pricing, underwriting, or risk classification priorities before committing to a full pricing review.

CORE BUSINESS QUESTION

Which segments require pricing, underwriting, or data quality review, and is there enough technical support to prioritize the next action?

What the diagnostic reviews

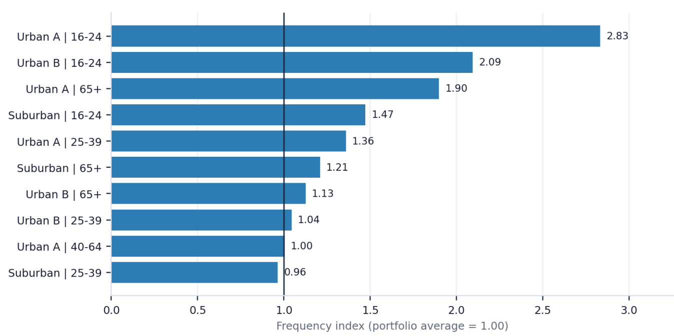
- Portfolio experience by segment
- Exploratory claim frequency patterns
- Benchmark Poisson frequency GLM with exposure offset
- Observed-versus-expected diagnostics
- Calibration by expected-risk decile

What it helps decide

- Where pricing pressure may exist
- Which underwriting rules, coverage terms, or risk classifications deserve review
- Where model calibration may be weak
- Which areas should be monitored first
- Whether deeper pricing work is justified

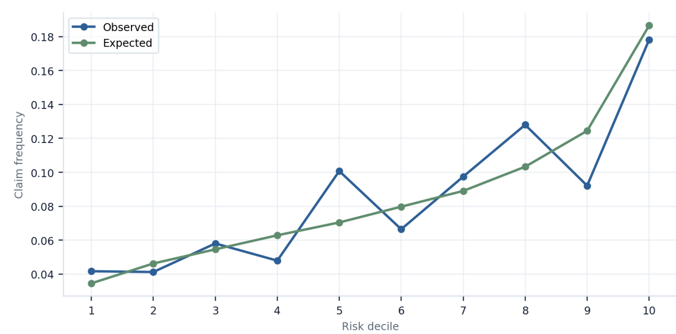
Elevated frequency segments

Frequency index compares each segment's observed claim frequency against the portfolio average. A value above 1.00 indicates higher frequency than the portfolio average.



Model calibration view

Policies are sorted by model-expected frequency and grouped into ten equally sized risk deciles. Higher deciles should generally show higher observed frequency.



Typical deliverables

Executive memo, Excel review pack, segment diagnostics, model validation summary, and technical appendix if needed.

Intended users

Pricing actuaries, pricing managers, underwriting managers, analytics teams, MGAs, program administrators, and consultants.

Suggested next step

Review one book, program, territory cluster, or segment group and identify 3–5 pricing or underwriting priorities.